Example 3.8

Find the value of $(3a+4c)^2$ by using $(a+b)^2$ identity.

Solution:

Comparing $(3a+4c)^2$ with $(a+b)^2$, we have $a=3a, b=4c$

Now $(a+b)^2 = a^2+2ab+b^2$

$\therefore \quad (3a+4c)^2 = (3a)^2 + 2(3a)(4c) + (4c)^2$ (replacing a and b values)

$= 3^2a^2 + (2\times3\times4)(a\times c) + 4^2c^2$

$(3a+4c)^2 = 9a^2 + 24ac + 16c^2$

Example 3.9

Find the value of 998^2 by using $(a-b)^2$ identity.

Solution:

We know, 998 can be expressed as $(1000-2)$

$\therefore \ (998)^2 = (1000-2)^2$

This is in the form of $(a-b)^2$, we get $a=1000, \ b=2$

Now $(a-b)^2 = a^2-2ab+b^2$

$(1000-2)^2 = (1000)^2 - 2(1000)(2) + (2)^2$

$(998)^2 = 1000000 - 4000 + 4 = 996004$

Think

Which is corrcet?
$(3a)^2$ is equal to
(i) $3a^2$ (ii) 3^2a
(iii) $6a^2$ (iv) $9a^2$

Example 3.10

Simplify $(3x+5y)(3x-5y)$ by using $(a+b)\ (a-b)$ identity.

Solution:

We have $(3x+5y)(3x-5y)$

Comparing it with $(a+b)(a-b)$ we get $a=3x \quad b=5y$

Now $(a+b)(a-b) = a^2-b^2$

$(3x+5y)(3x-5y) = (3x)^2 - (5y)^2$ (replacing a and b values)

$= 3^2x^2 - 5^2y^2$

$(3x+5y)(3x-5y) = 9x^2 - 25y^2$

Example 3.11

Expand y^2-16 by using a^2-b^2 identity

Solution:

y^2-16 can be written as y^2-4^2

Comparing it with a^2-b^2, we get $a=y, b=4$

Now $a^2-b^2 = (a+b)(a-b)$

$y^2-4^2 = (y+4)(y-4)$

$y^2-16 = (y+4)(y-4)$

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